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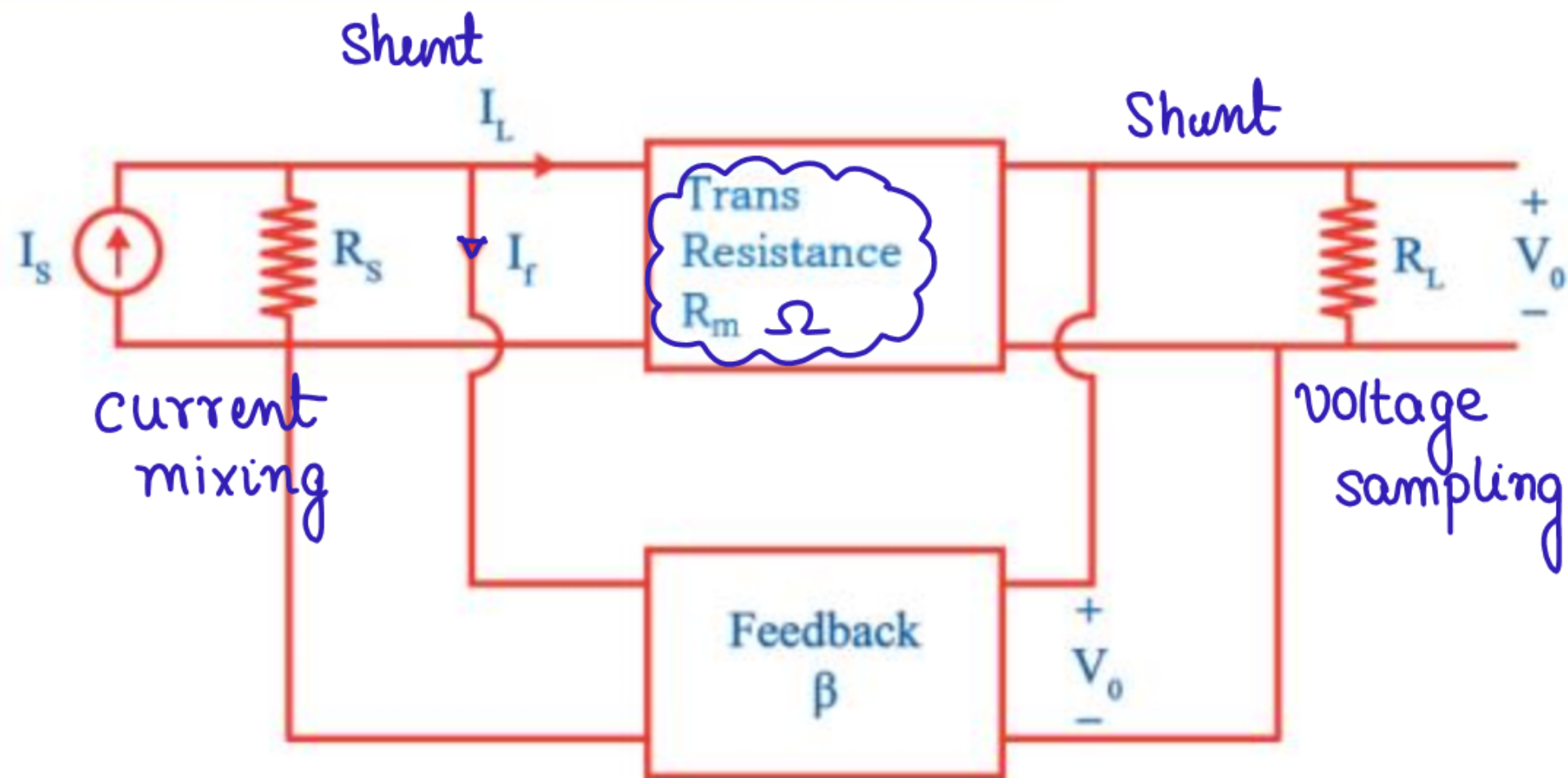


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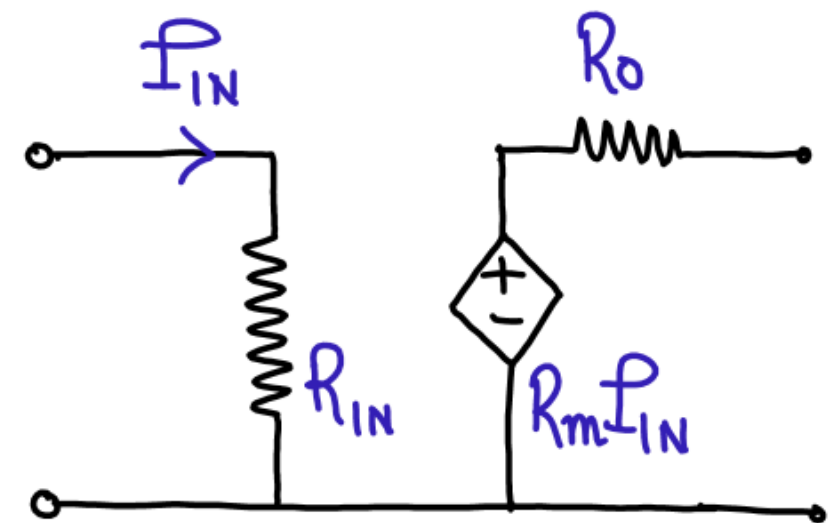
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(2) Shunt - Shunt Feedback

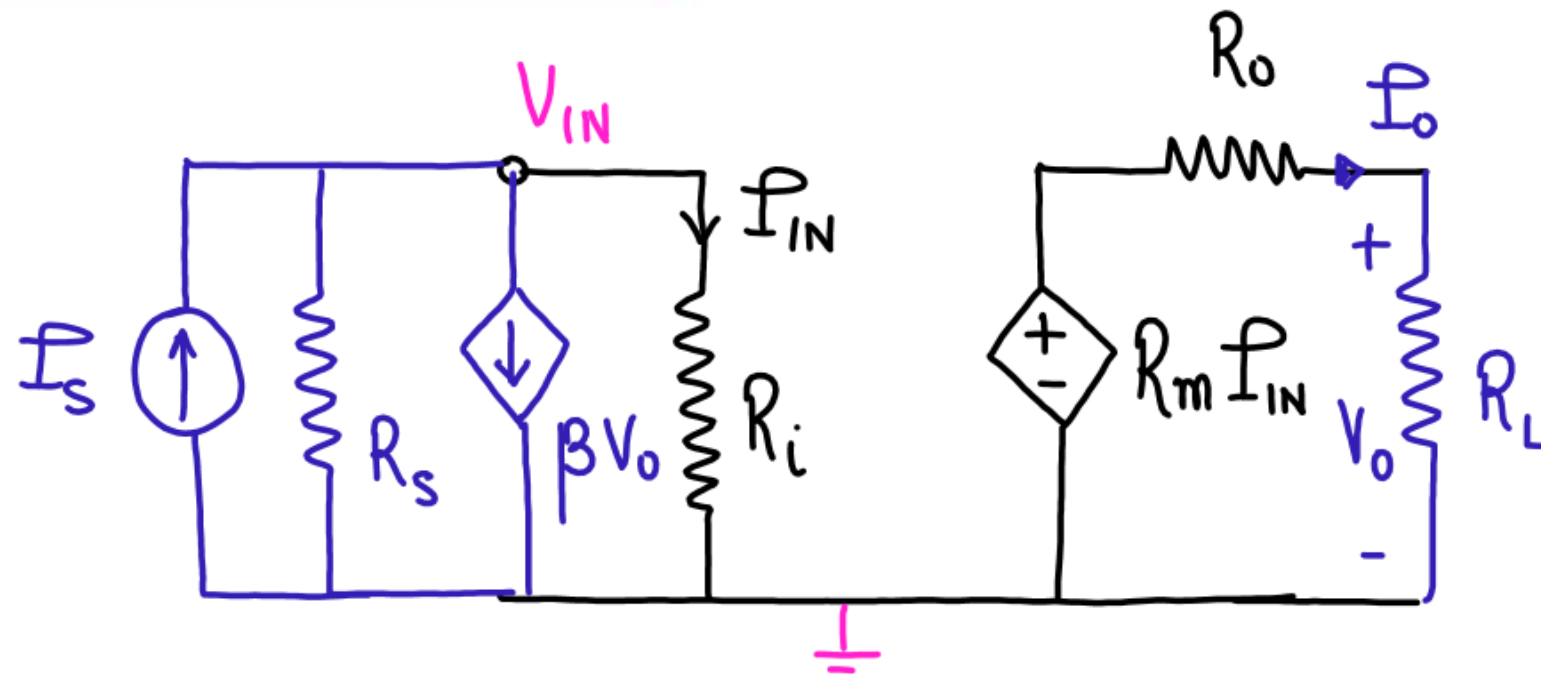


$$I_f = \beta V_o$$

$\beta: v$



Equivalent Circuit



$$I_s - \beta V_o = \frac{V_{IN}}{R_s} + \frac{V_{IN}}{R_i}$$

$$V_o = R_m I_{IN} \times \frac{R_L}{R_o + R_L} \\ = R_{ML} I_{IN}$$

$$R_{ML} = R_m \cdot \frac{R_L}{R_o + R_L}$$

$$I_s - \beta R_{ML} I_{IN} = \frac{V_{IN}}{R_s} + \frac{V_{IN}}{R_i}$$

$$I_s = \frac{V_{IN}}{R_s} + \frac{V_{IN}}{R_i} (1 + \beta R_{ML})$$

$$R_{if} = \frac{R_i}{1 + \beta R_{ML}} < R_i$$

$$V_o = R_m I_{IN} - I_o R_o$$

$$I_{IN} = \frac{(I_s - \beta V_o) \times R_s}{R_s + R_{IN}}$$

$$V_o = R_m \times \frac{R_s}{R_s + R_{IN}} (I_s - \beta V_o) - I_o R_o$$

$$= R_{ms} I_s - R_{ms} \beta V_o - I_o R_o$$

$$V_o (1 + R_{ms} \beta) = R_{ms} I_s - I_o R_o$$

$$V_o = \frac{R_{ms}}{1 + R_{ms} \beta} I_s - \frac{I_o R_o}{1 + R_{ms} \beta}$$

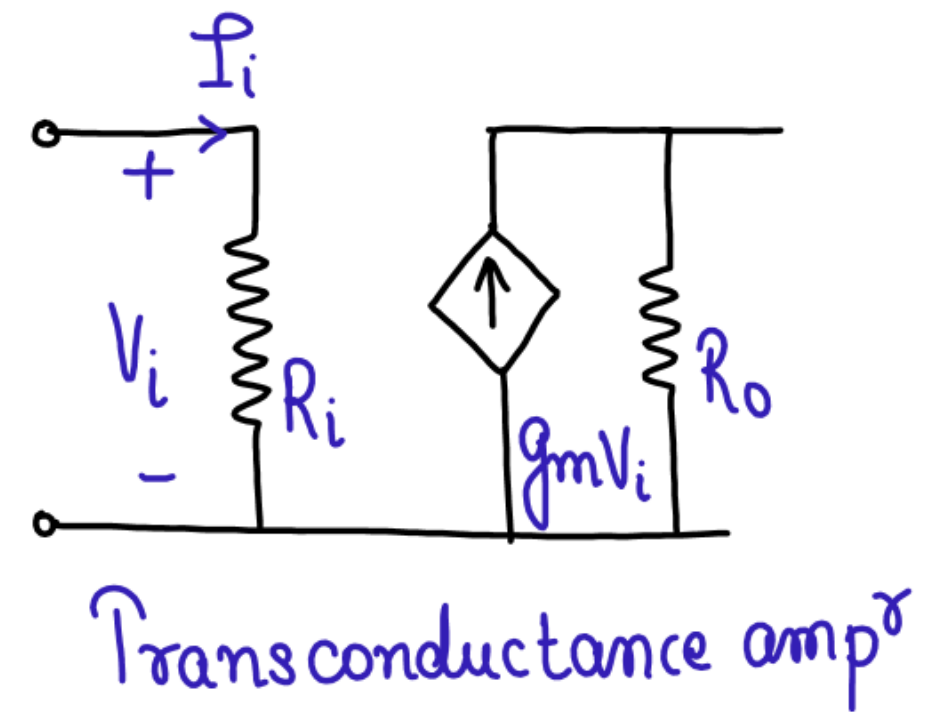
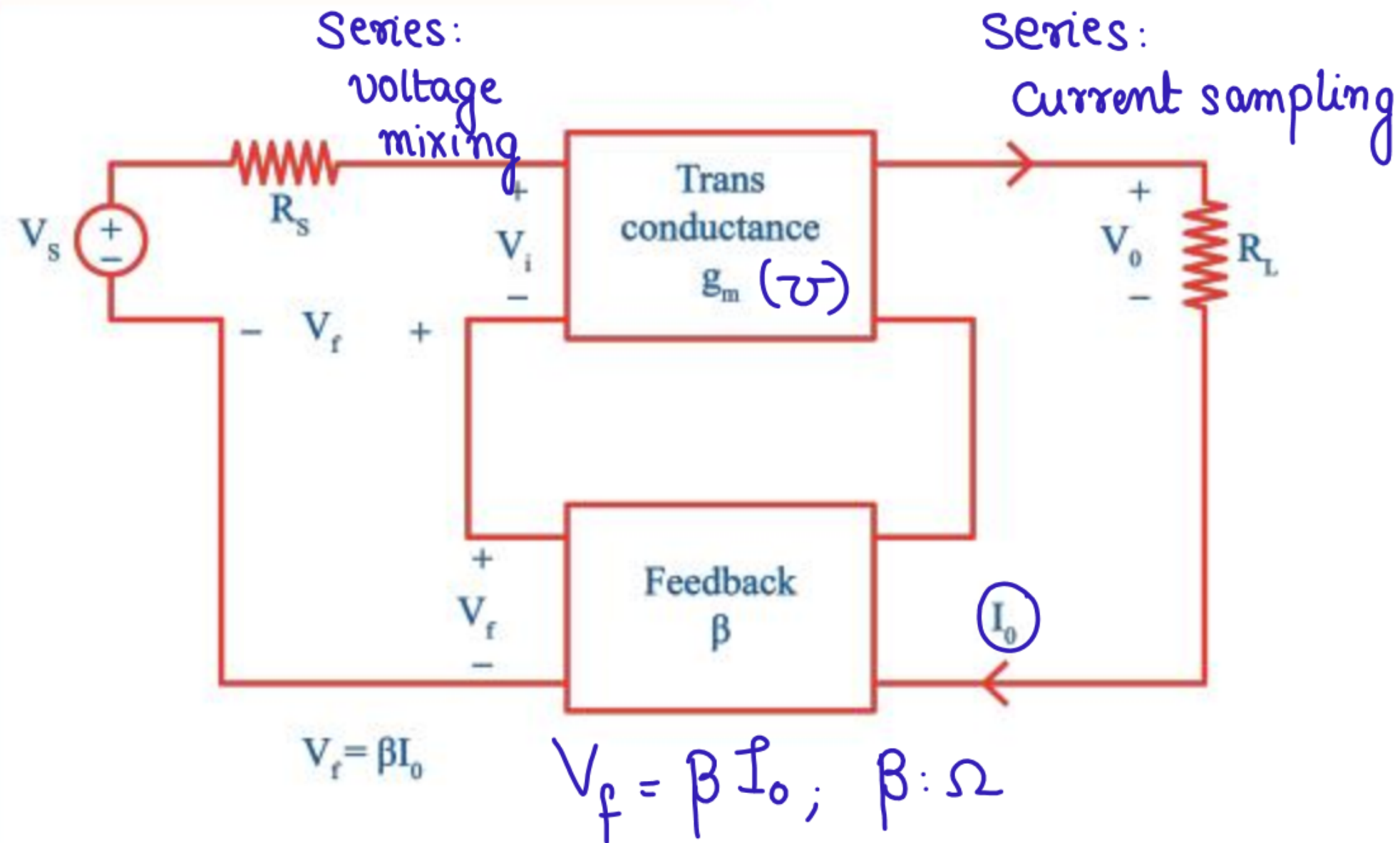
o/p resistance with
feedback

$$R_{of} = \frac{R_o}{1 + R_{ms} \beta} < R_o$$

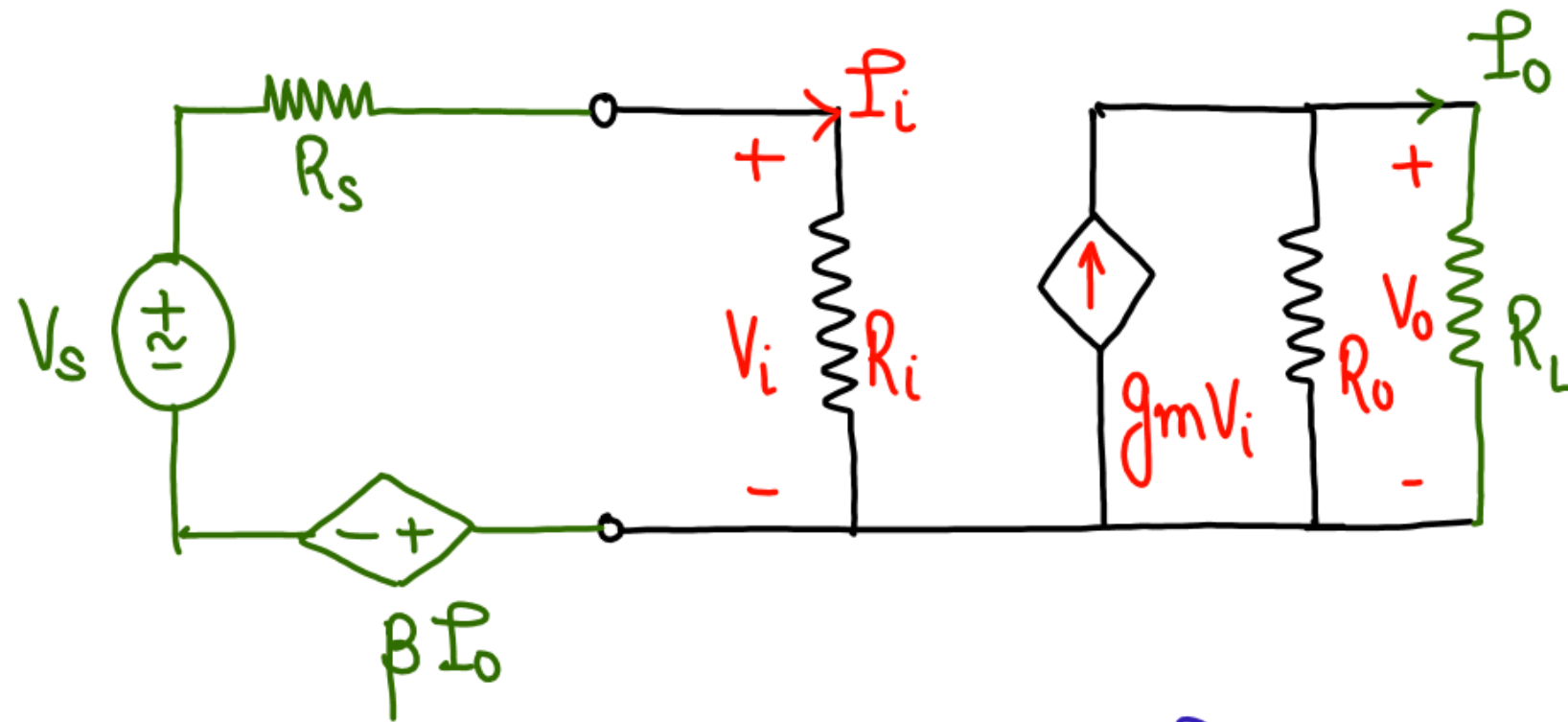
$$R_{ms} = R_m \times \frac{R_s}{R_s + R_i}$$

(3) Current - Series Negative Feedback

Series- Series Feedback



Transconductance Amplifier



$$V_s - \beta I_o = I_i R_s + I_i R_i$$

$$I_o = g_m V_i \frac{R_o}{R_o + R_L} = g_{mL} V_i$$

$$g_{mL} = g_m \frac{R_o}{R_o + R_L}$$

$$\begin{aligned} V_s &= I_i R_s + I_i R_i + \beta g_{mL} V_i \\ &= I_i R_s + I_i R_i (1 + g_{mL} \beta) \end{aligned}$$

input resistance with f/b

$$R_{if} = R_i (1 + g_{mL} \beta) > R_i$$

o/p side

$$I_o = g_m V_i - V_o / R_o$$

$$V_i = (V_s - \beta I_o) \frac{R_i}{R_s + R_i}$$

$$\begin{aligned} I_o &= g_m \frac{R_i}{R_s + R_i} (V_s - \beta I_o) - V_o / R_o \\ &= g_{ms} V_s - g_{ms} \beta I_o - V_o / R_o \end{aligned}$$

$$I_o (1 + g_{ms} \beta) = g_{ms} V_s - V_o / R_o$$

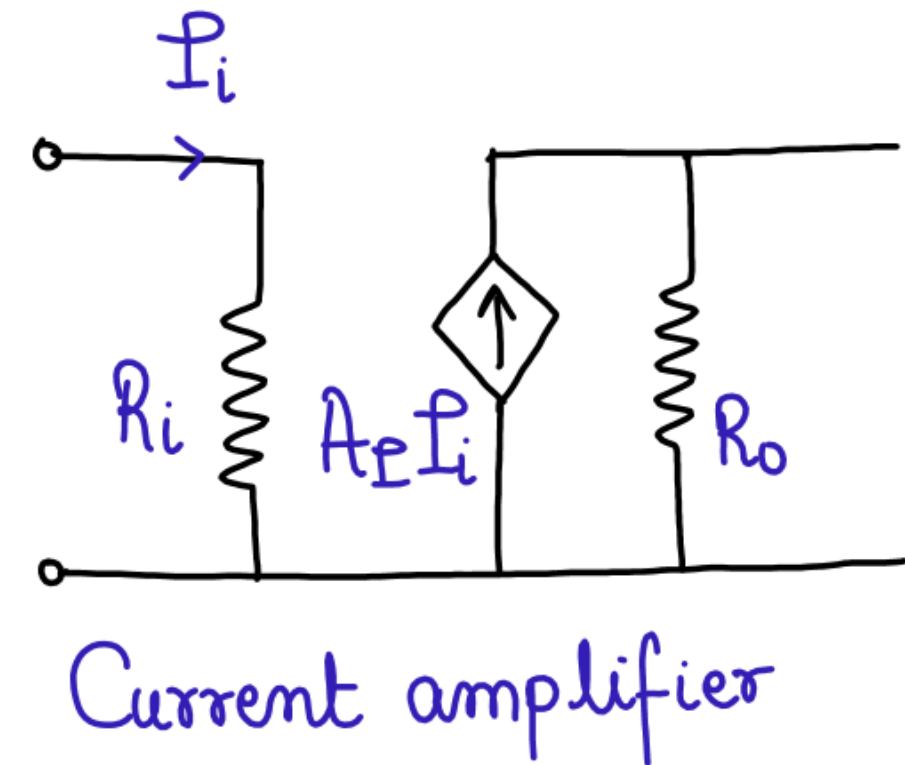
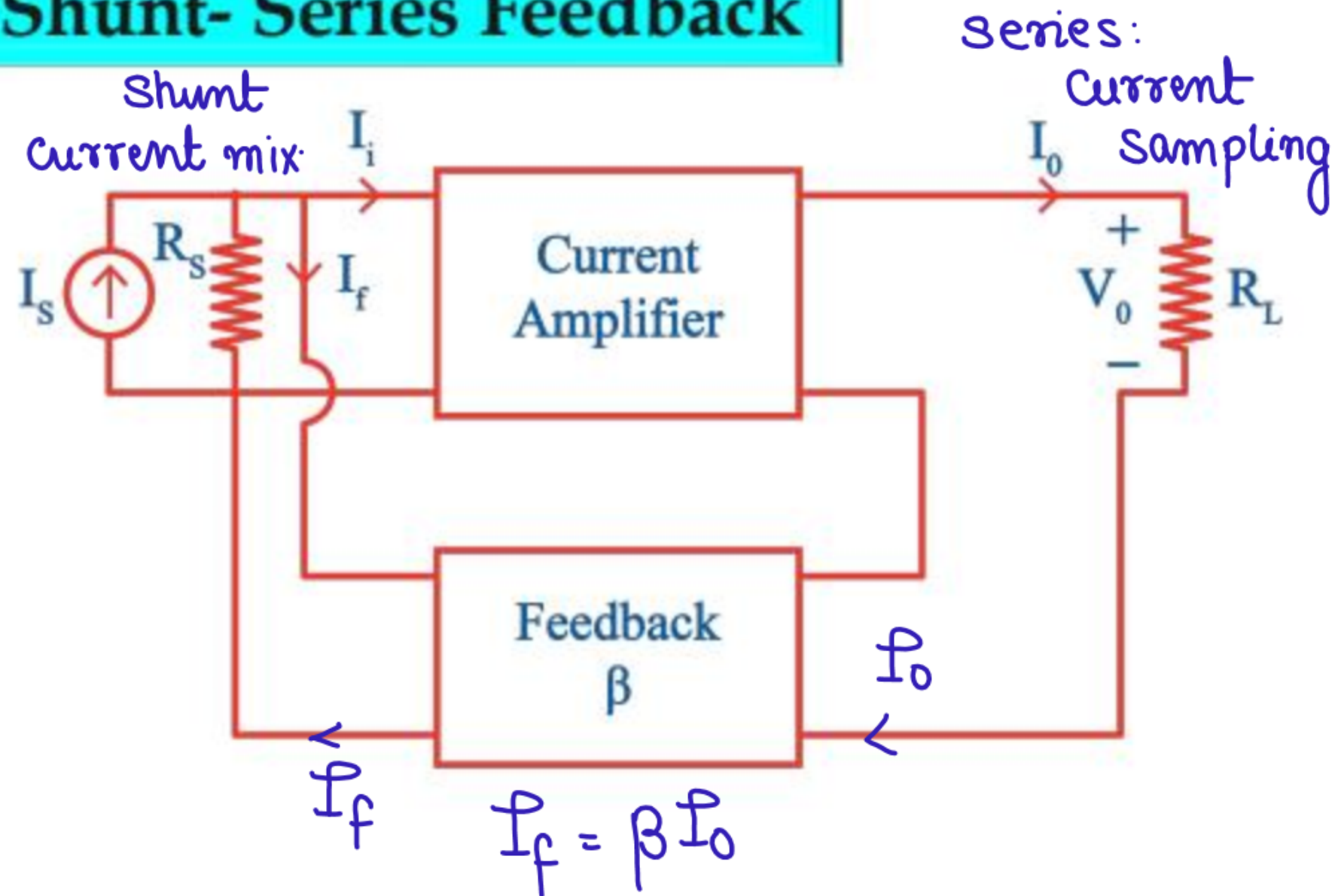
$$I_o = \frac{g_{ms}}{1 + g_{ms} \beta} V_s - \frac{V_o}{R_o (1 + g_{ms} \beta)}$$

o/p resistance with f/b

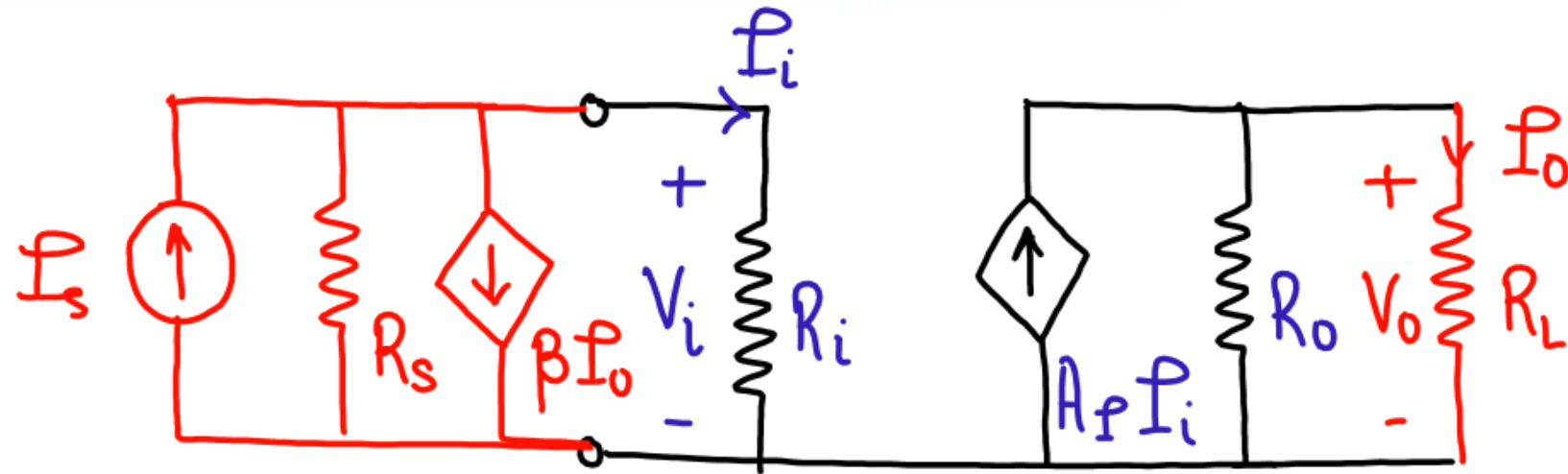
$$R_{of} = R_o (1 + g_{ms} \beta) > R_o$$

(4) Current - Shunt Negative Feedback

Shunt-Series Feedback



Circuit of Current Amplifier



$$I_s - \beta I_o = V_i/R_s + V_i/R_i$$

$$I_o = A_f I_i \times \frac{R_o}{R_o + R_L} = A_{fL} I_i$$

$$A_{fL} = A_f \frac{R_o}{R_o + R_L}$$

$$I_s - A_{fL} \beta I_i = V_i/R_s + V_i/R_i$$

$$I_s = \frac{V_i}{R_s} + \frac{V_i}{R_i} [1 + A_{fL} \beta]$$

input resistance with f/b

$$R_{if} = \frac{R_i}{1 + A_{fL} \beta} < R_i$$

$$I_o = A_{fs} I_i - V_o / R_o$$

$$I_o = (I_s - \beta I_o) \times \frac{R_s}{R_s + R_i}$$

$$= A_{fs} \times \frac{R_s}{R_s + R_i} (I_s - \beta I_o) - V_o / R_o$$

$$= A_{fs} I_s - A_{fs} \beta I_o - V_o / R_o$$

$$I_o (1 + A_{fs} \beta) = A_{fs} I_s - V_o / R_o$$

$$I_o = \frac{A_{fs}}{1 + A_{fs} \beta} I_s - \frac{V_o}{R_o (1 + A_{fs} \beta)}$$

o/p resistance with f/b

$$R_{of} = R_o (1 + A_{fs} \beta) > R_o$$

Summary

Source	Amplifier	Feedback	Feedback	R_{if}	R_{of}
VCVS	Voltage series amp ^r	Series-shunt	Voltage series	$R_i(1+A\beta)$	$\frac{R_o}{1+A\beta}$
VCCS	Trans. Conductance	Series-Series	Current series	$A: g_m$ $R_i(1+A\beta)$	$R_o(1+A\beta)$
CCVS	Trans. Resistance	Shunt-shunt	Voltage series shunt	$A: R_m$ $R_i / (1+A\beta)$	$R_o / (1+A\beta)$
CCCS	Current	Shunt-Series	Current series shunt	$\frac{R_i}{1+A\beta}$	$R_o(1+A\beta)$

How to identify feedback?

- draw circuit of amp^r
- check KCL & KVL at i/p terminal

$$I_i = I_s$$

$$V_i = V_s$$

V_i : contains f/b

I_i : contain f/b

voltage mixing

current mixing

- current mixing : $V_i = 0$ compute I_f

$$I_f = f(I_o) \text{ or } f(V_o) \leftarrow \text{must not contain } R_L$$

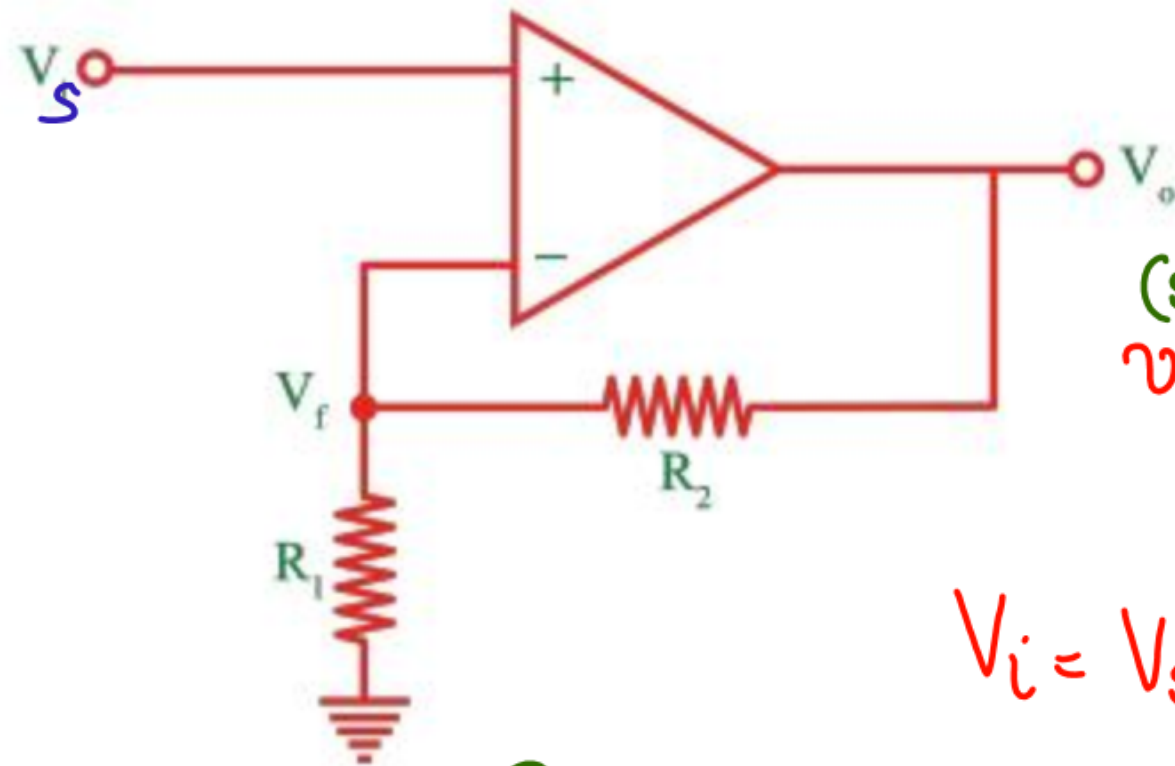
voltage mixing : $I_i = 0$ compute V_f

$$V_f = f(I_o) \text{ or } f(V_o) \leftarrow \text{must not contain } R_L$$



Question-01

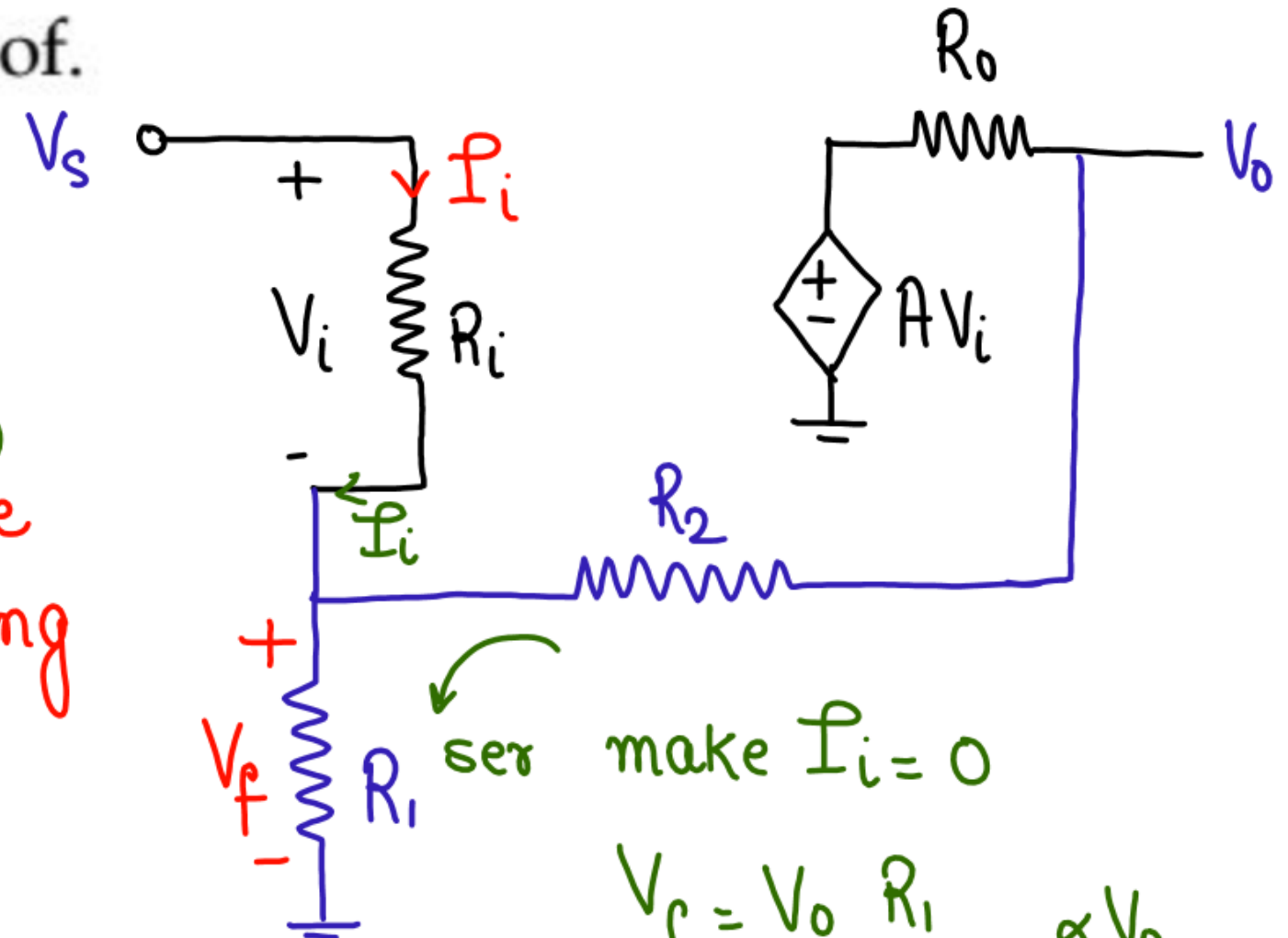
The following circuit is an example of.



(Series)
voltage
mixing

$$V_i = V_s - V_f$$

Series-Shunt f/b



ser make $I_i = 0$

$$V_f = V_o \frac{R_1}{R_1 + R_2} \propto V_o$$

voltage-sampling
(shunt)

$$\beta = \frac{R_1}{R_1 + R_2}$$

$$\text{gain} = \frac{A}{1 + A\beta}$$
$$= \frac{1}{\frac{1}{A} + \beta}$$

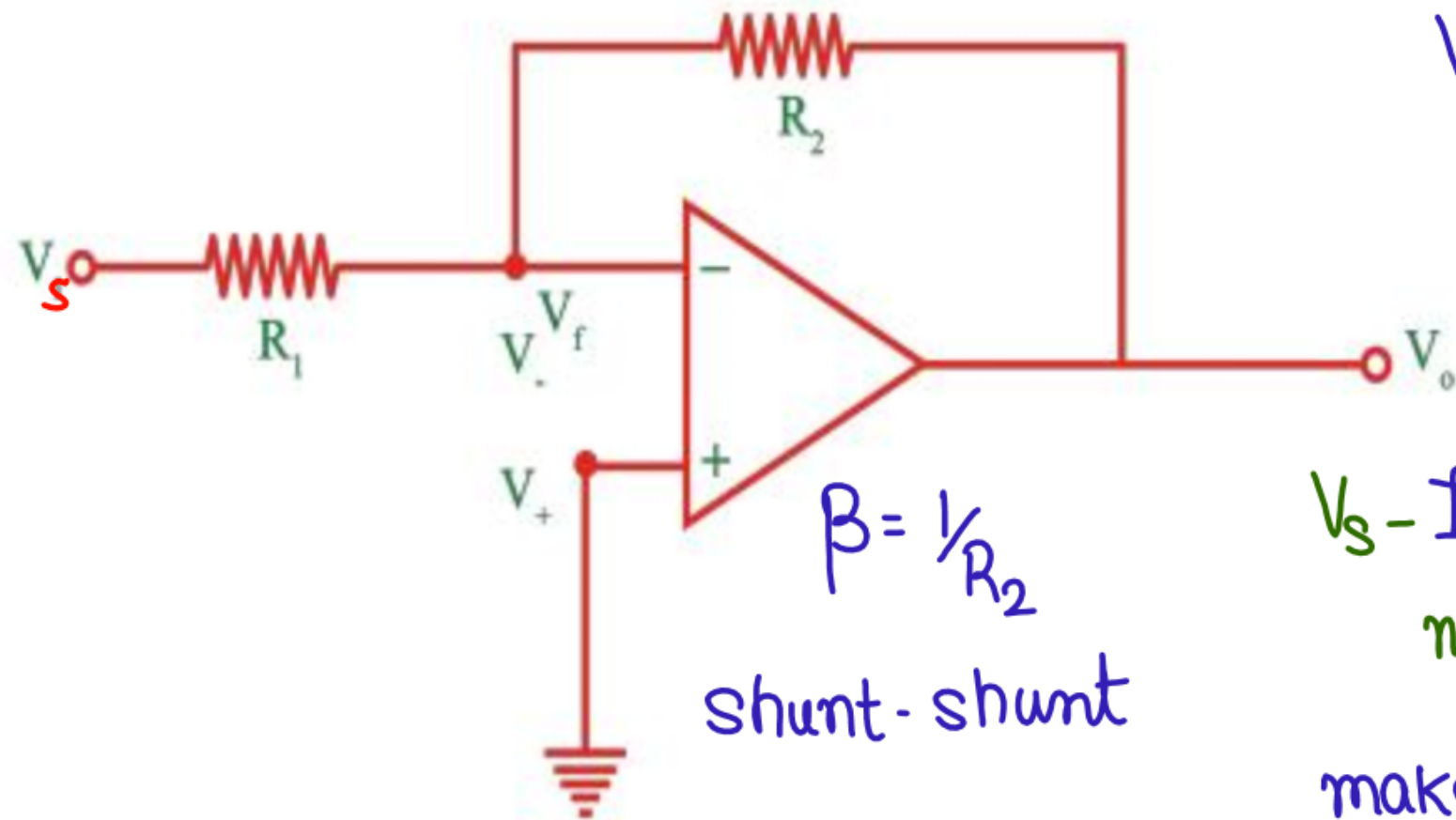
if open loop gain $\rightarrow \infty$

$$\text{gain} = \frac{1}{\beta} = \frac{R_1 + R_2}{R_1}$$
$$= 1 + R_2/R_1$$



Question-02

The following circuit is an example of ?



$\beta = 1/R_2$
shunt-shunt

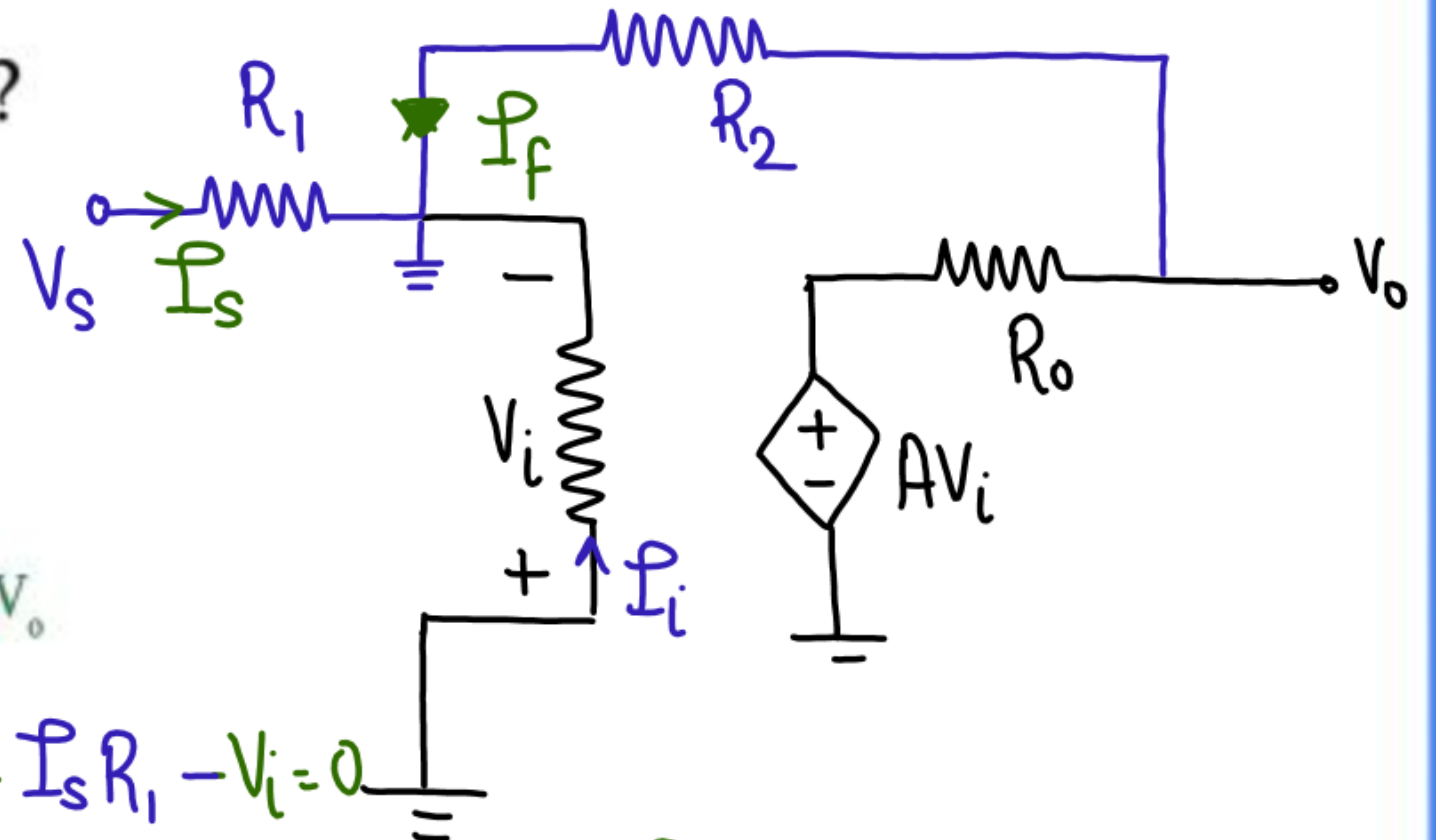
$$V_s - I_s R_1 - V_i = 0$$

no f/b

make $V_i = 0$

$$I_f = V_o / R_2 \propto V_o$$

voltage sampling



$$I_s + I_f + I_i = 0$$

$$I_i = -I_s - I_f$$

current mixing



Question-03

An amplifier has open loop gain of 100 and its lower and upper cutoff frequencies are 100Hz and 100kHz respectively. A feedback network with a feedback factor of 0.99 is connected to amplifier. Find lower and upper cut off frequencies.

$$1 + A\beta = 1 + 100 \times 0.99 \\ = 100$$

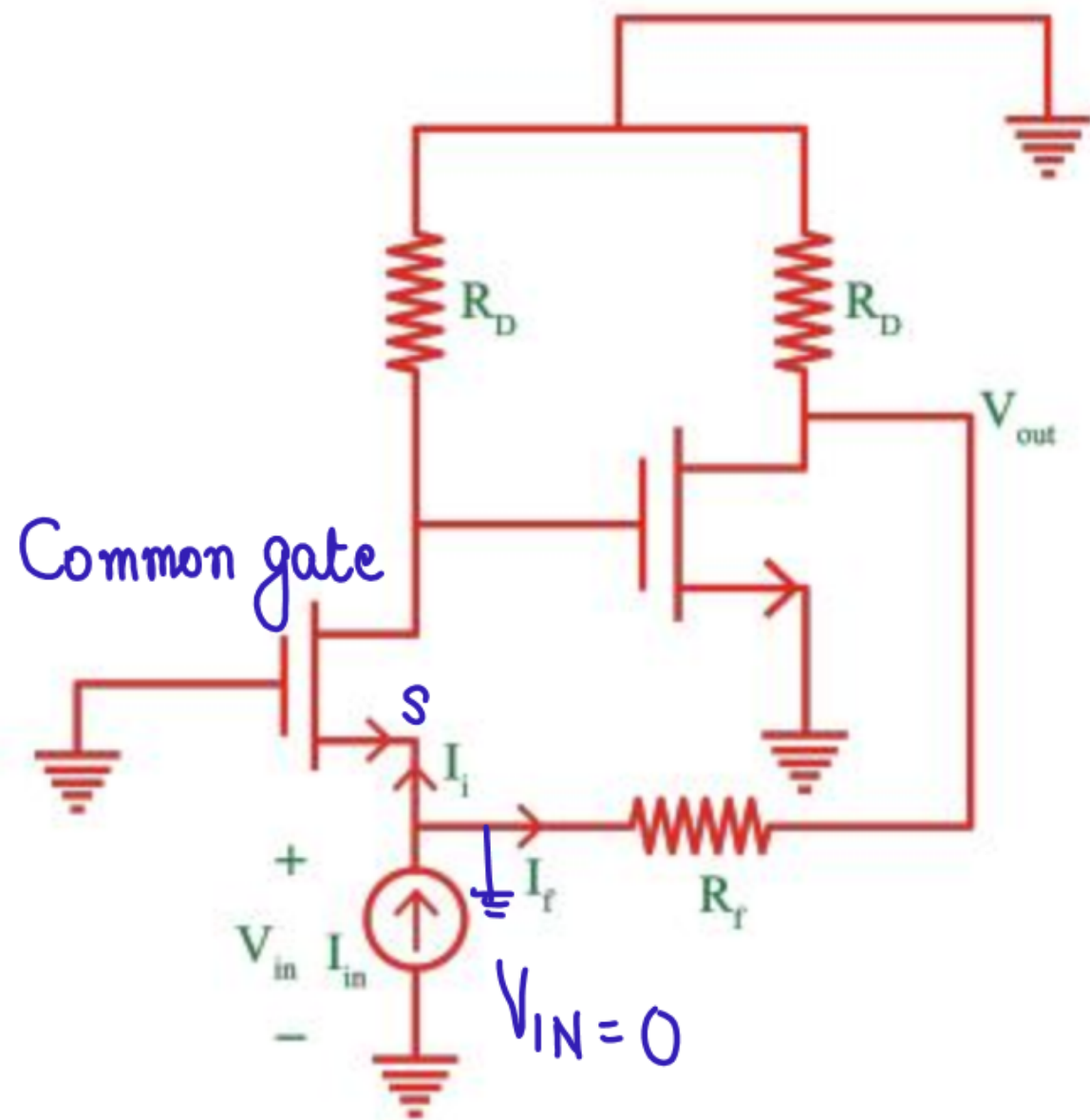
$$f_L = \frac{f_L}{1 + A\beta} = \frac{100}{100} = 1 \text{ Hz}$$

$$f_H = f_H (1 + A\beta) = 10 \text{ MHz}$$



Question-04

In the ac equivalent circuit shown in the figure if I_{in} is the input current and R_f is very large then find the nature of feedback .



$$I_i = I_s - I_f : \text{current mixing (shunt)}$$

$$I_f = \frac{0 - V_{out}}{R_f} \propto V_{out} ; \beta = -1/R_f$$

voltage sampling (shunt)

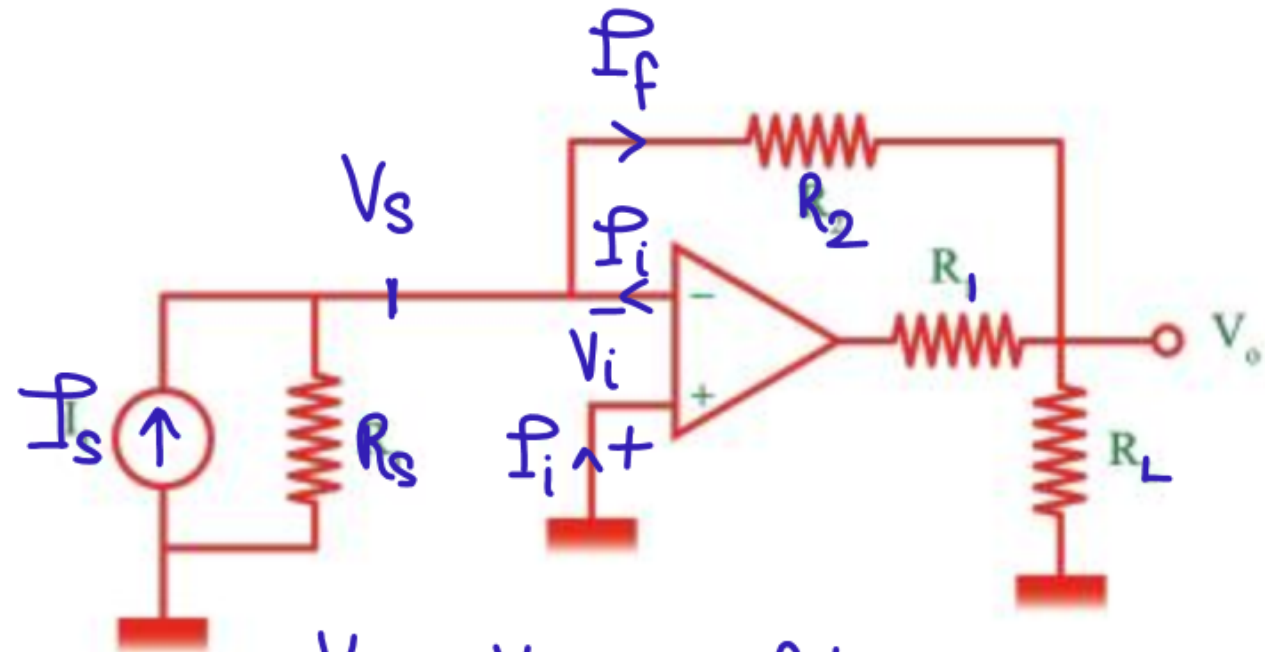
Shunt-shunt f/b



Question-05

The circuit shown in the figure is an example of

- a. series – shunt feedback
- b. shunt – series feedback
- ✓ c. shunt – shunt feedback
- d. series – series feedback



$$V_i = -V_s : \text{no f/b}$$

Current mixing

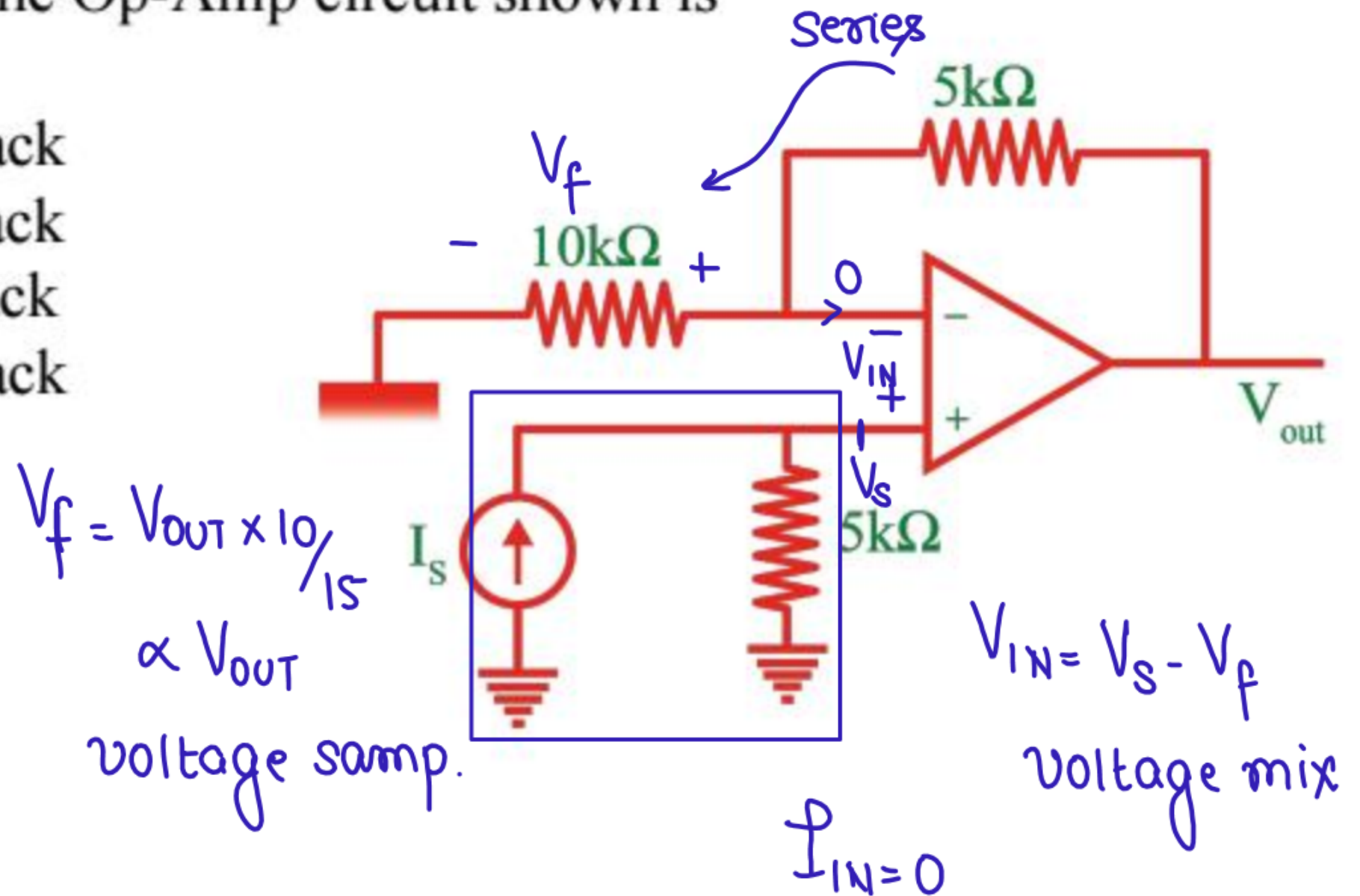
$$V_i = 0 ; I_f = \frac{0 - V_o}{R_2} \propto V_o \text{ voltage sampling}$$



Question-06

The nature of feedback in the Op-Amp circuit shown is

- samp. mixing*
- a. Current – Current feedback
 - ☒ b. Voltage – Voltage feedback
 - c. Current – voltage feedback
 - d. Voltage – Current feedback





Answers

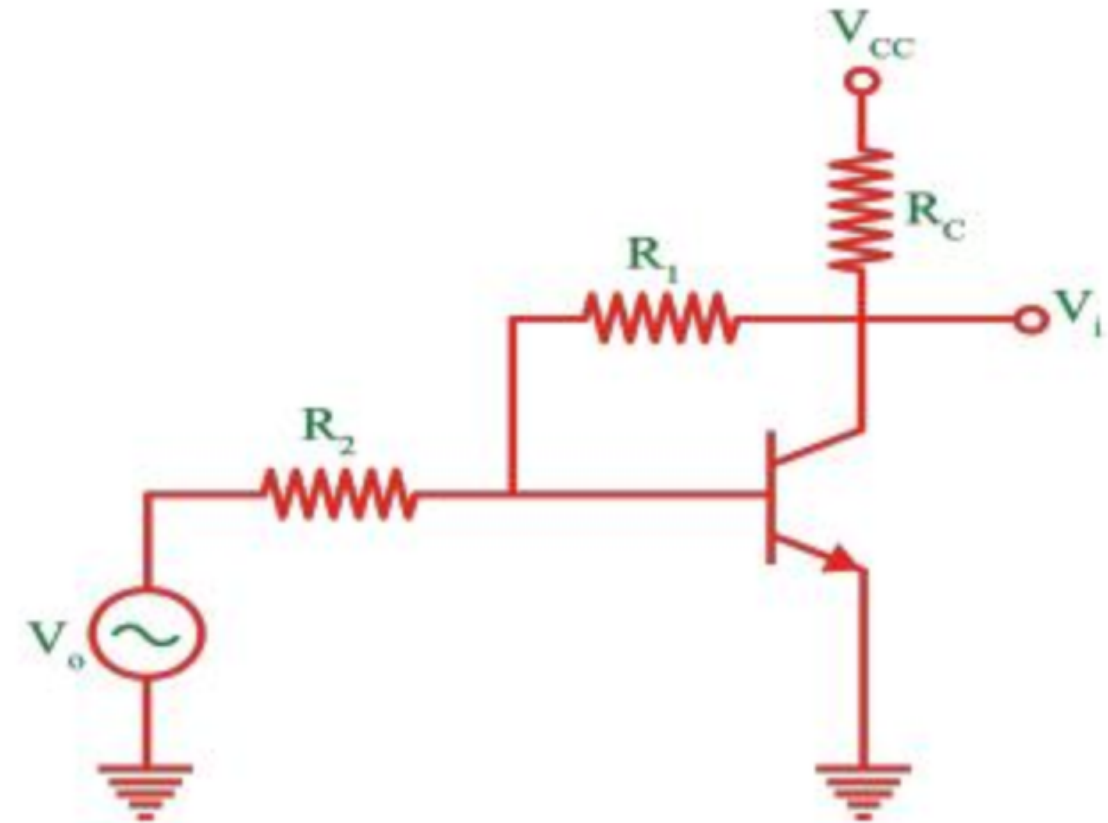
Question No.	Answer
1	Series Shunt
2	Shunt-Shun
3	10Mhz
4	100k Ω , 1 Ω
5	Shunt – Shunt feedback
6	B



Student's Assignment

Question-01

In the figure shown below



The feedback configuration and the approximate gain of the circuit is

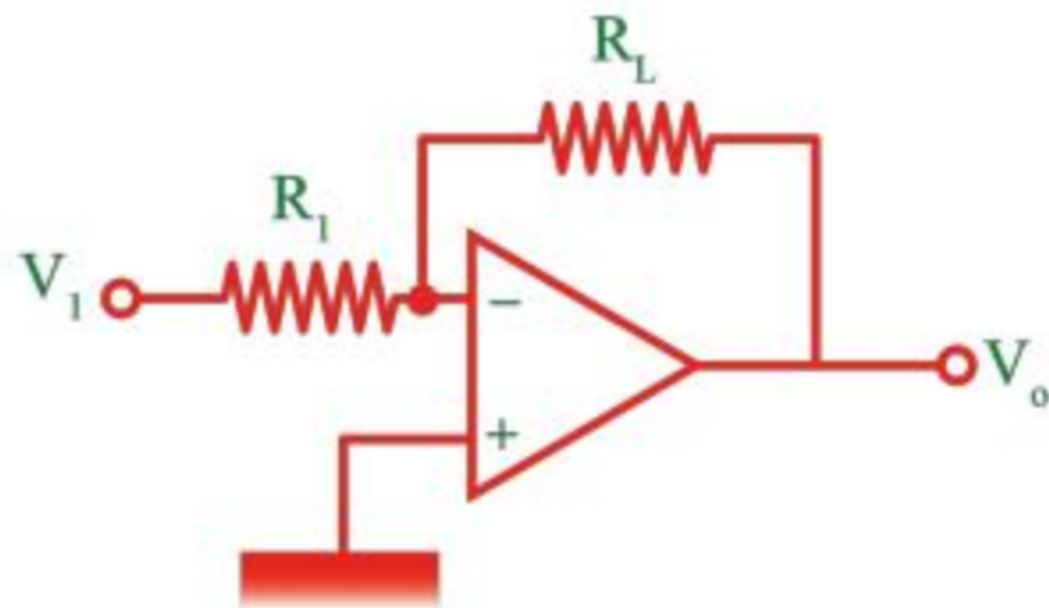
a. Shunt-Shunt, $A_v = -\frac{R_2}{R_1}$
c. Series-shunt, $A_v = -\frac{R_1}{R_1}$

b. shunt-series, $A_v = \frac{R_2}{R_1}$
d. Shunt-shunt, $A_v = -\frac{R_1}{R_C}$



Question-02

The following circuit is an example of



- a. Series – shunt feedback amplifier
- b. Shunt – series feedback amplifier
- c. Series – series feedback amplifier
- d. Shunt – shunt feedback amplifier



Question-03

A common Emitter Amplifier with R_E and without by-pass capacitor is also a

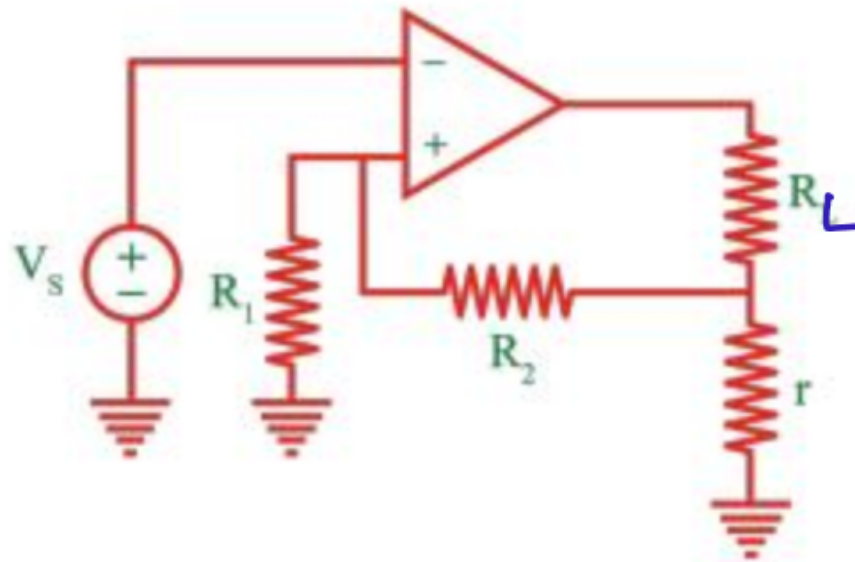
- (a) Voltage series negative feedback amplifier
- (b) Voltage shunt negative feedback amplifier
- (c) Current series negative feedback amplifier
- (d) Current shunt negative feedback amplifier



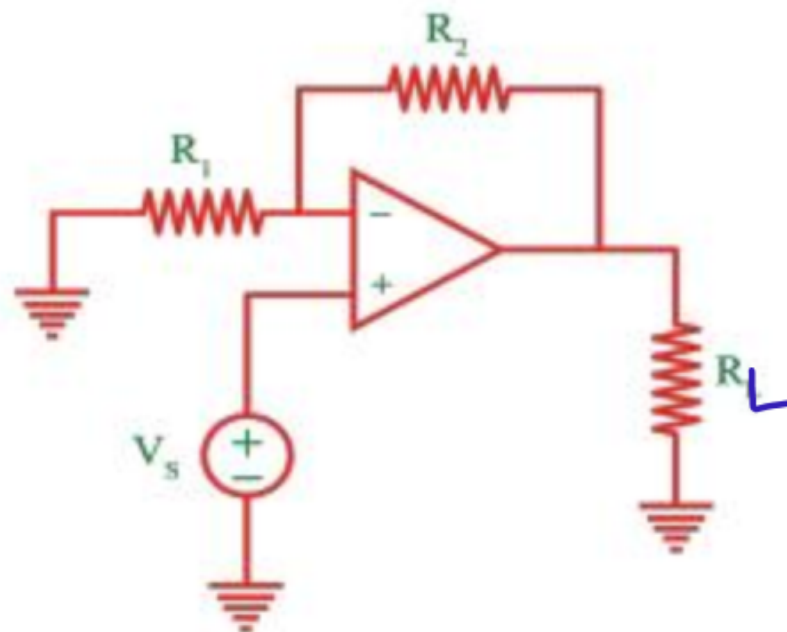
Question-04

Match List – I with List – II
LIST – I

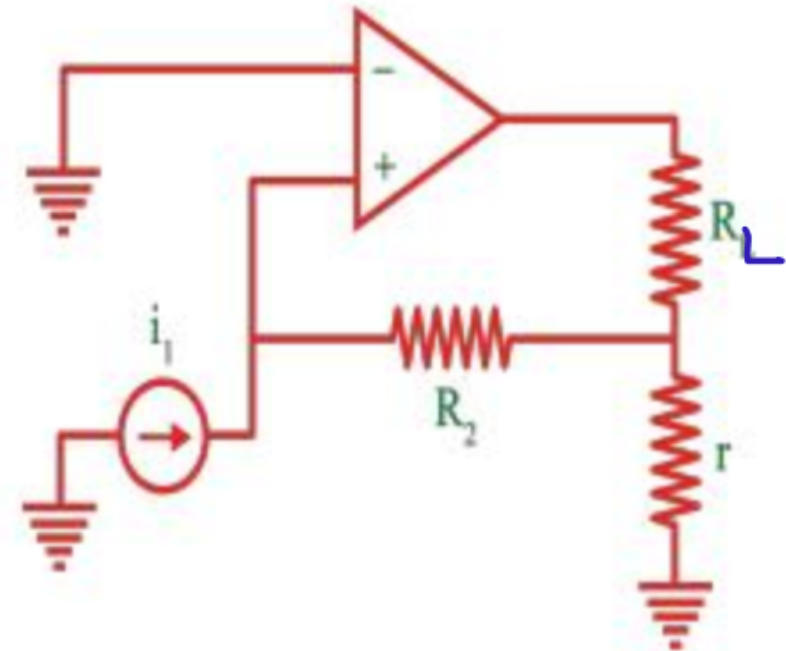
(a)



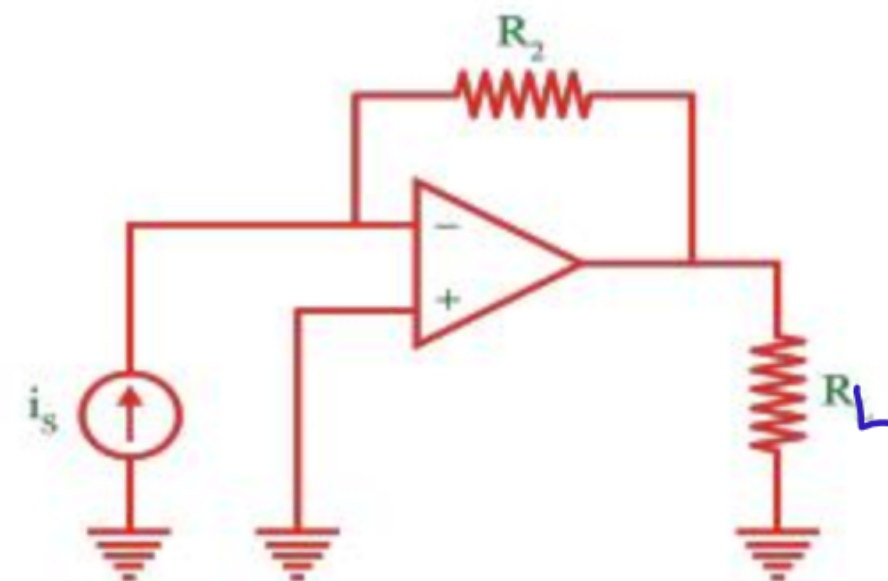
(c)



(b)



(d)





Question-04

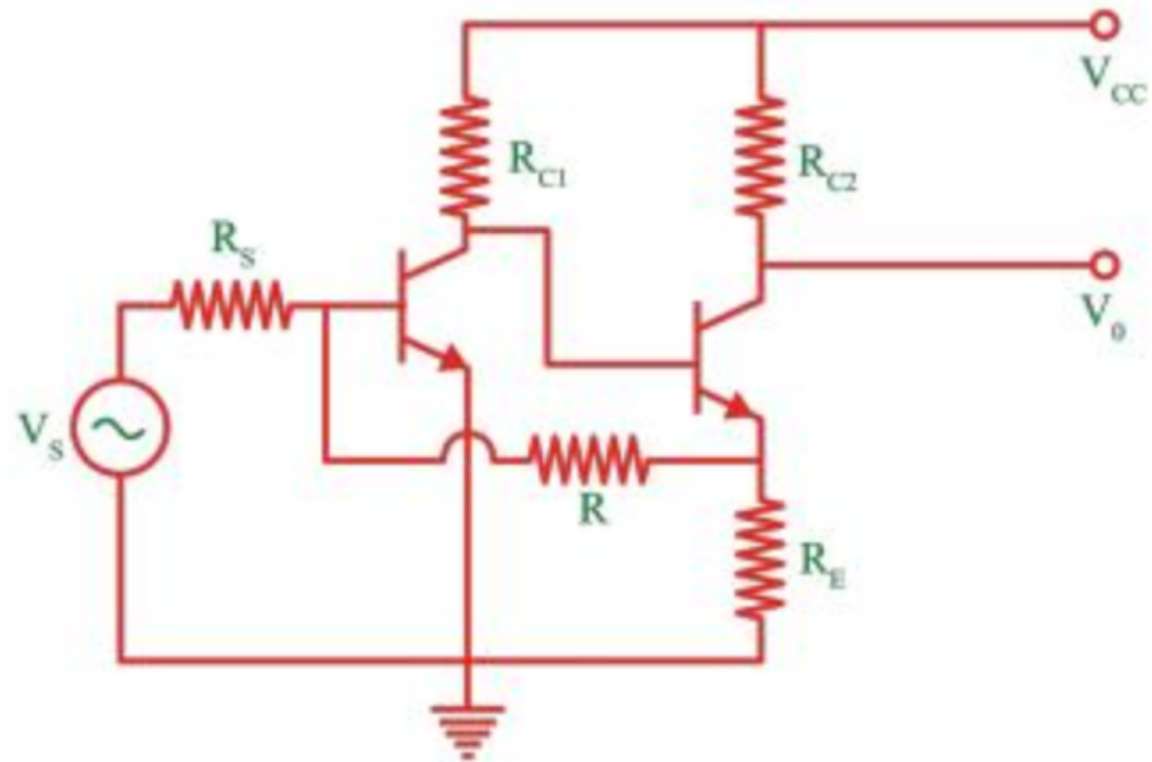
LIST – II

1. Shunt - Shunt
 2. Series – Shunt
 3. Series – Series
 4. Shunt – Series
- (a) A – 1, B – 2, C – 3, D – 4
 - (b) A – 2, B – 1, C – 4, D – 3
 - (c) A – 2, B – 1, C – 3, D – 4
 - (d) A – 4, B – 3, C – 2, D – 1



Question-05

In the circuit shown , the feedback type is

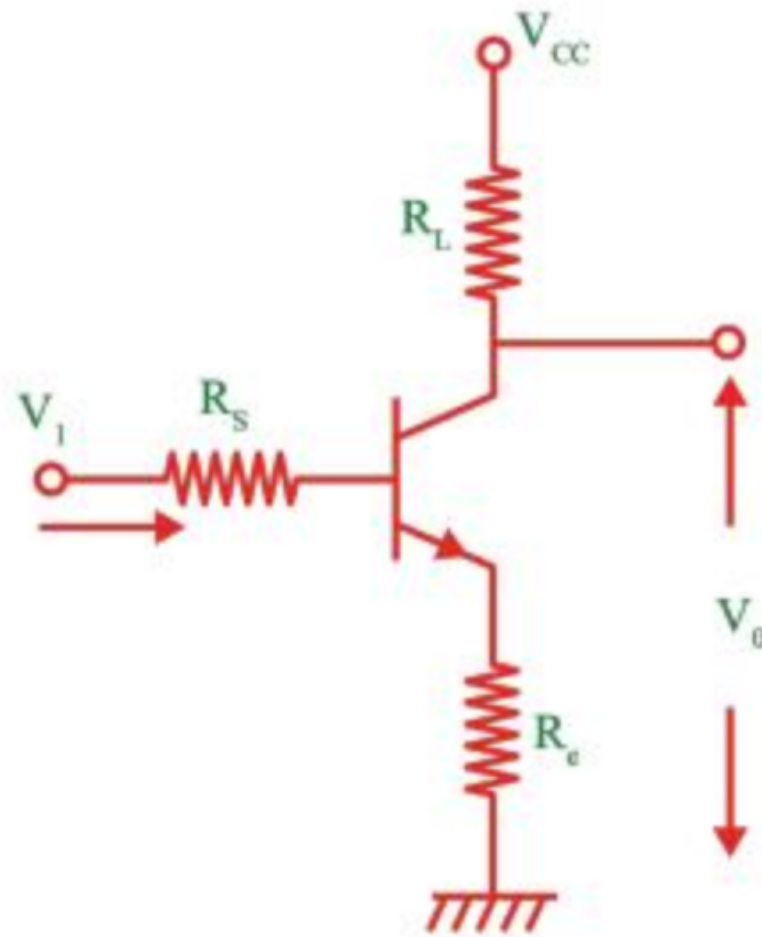




Question-06

Circuit of a feedback amplifier having series type feedback is shown in the given figure. The β of feedback network is determined by

- (a) $-R_e/R_L$
- (b) $-R_e$
- (c) $-R_L/R_e$
- (d) $-(R_e/-R_L)2$

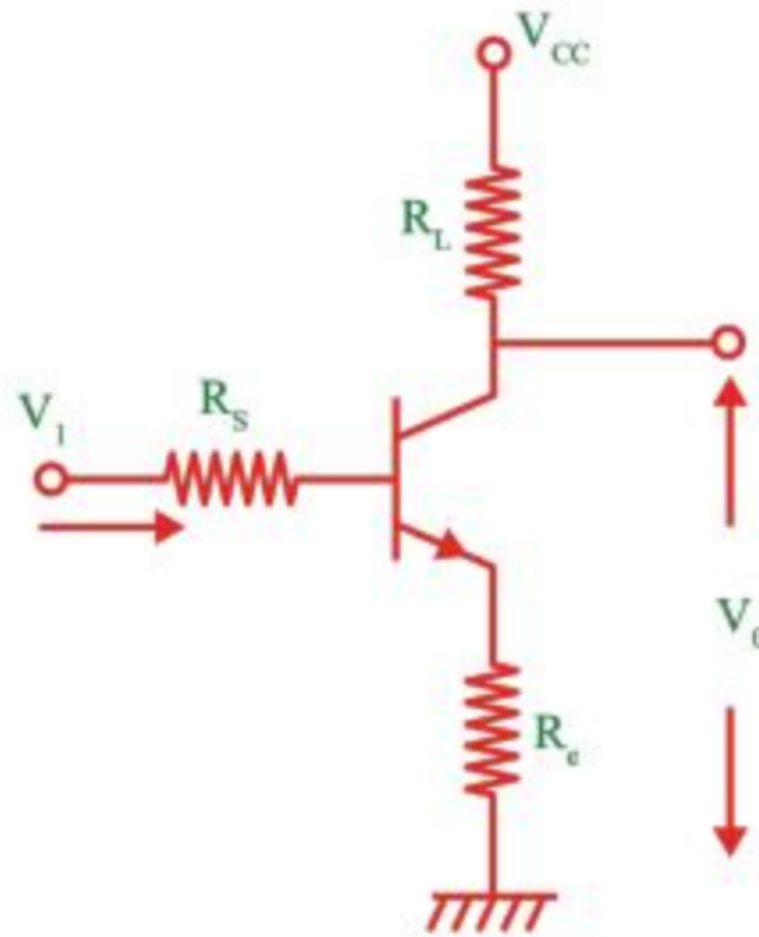




Question-06

Circuit of a feedback amplifier having series type feedback is shown in the given figure. The β of feedback network is determined by

- (a) $-R_e/R_L$
- (b) $-R_e$
- (c) $-R_L/R_e$
- (d) $-(R_e/-R_L)2$





Answers

Question No.	Answer
1	A
2	D
3	C
4	C
5	Current – Shunt
6	B



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